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LAB 10

Program 1: BEFORE INSERT Trigger for Validation

Query/Input:

```
CREATE TABLE EMPLOYEE_VALIDATION (
   emp_id INT PRIMARY KEY,
   emp_name VARCHAR(50),
   salary DECIMAL(10,2)
);

DELIMITER //

CREATE TRIGGER BeforeEmployeeInsert
BEFORE INSERT ON EMPLOYEE_VALIDATION
FOR EACH ROW
BEGIN
   IF NEW.salary < 0 THEN
      SIGNAL SQLSTATE '45000'
      SET MESSAGE_TEXT = 'Salary cannot be negative';
   END IF;
END //

DELIMITER ;

INSERT INTO EMPLOYEE_VALIDATION
VALUES (1, 'Alice', 30000);
```

SELECT * FROM EMPLOYEE_VALIDATION;

Output:

Output:

```
+--------+----------+----------+
| emp_id | emp_name | salary   |
+--------+----------+----------+
|      1 | Alice    | 30000.00 |
+--------+----------+----------+
```

Program 2: AFTER INSERT Trigger to Record Employee Insertion

Query/Input:

```
CREATE TABLE EMPLOYEE_AUDIT (
   audit_id INT AUTO_INCREMENT PRIMARY KEY,
   action_type VARCHAR(20),
   emp_id INT,
   action_time TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

DELIMITER //

CREATE TRIGGER AfterEmployeeInsert
AFTER INSERT ON EMPLOYEE_VALIDATION
FOR EACH ROW
BEGIN
   INSERT INTO EMPLOYEE_AUDIT (action_type, emp_id)
   VALUES ('INSERT', NEW.emp_id);
END //

DELIMITER ;
```

```
INSERT INTO EMPLOYEE_VALIDATION
VALUES (2, 'Bob', 35000);

SELECT * FROM EMPLOYEE_AUDIT;
```

Output:

```
+----------+-------------+--------+---------------------+
| audit_id | action_type | emp_id | action_time         |
+----------+-------------+--------+---------------------+
|        1 | INSERT      |      2 | 2026-10-09 07:32:22 |
+----------+-------------+--------+---------------------+
```

Program 3: BEFORE UPDATE Trigger for Salary Validation

Query/Input:

```
DELIMITER //

CREATE TRIGGER BeforeEmployeeUpdate
BEFORE UPDATE ON EMPLOYEE_VALIDATION
FOR EACH ROW
BEGIN
  IF NEW.salary < 0 THEN
    SIGNAL SQLSTATE '45000'
    SET MESSAGE_TEXT = 'Salary cannot be negative';
  END IF;
END //

DELIMITER ;

UPDATE EMPLOYEE_VALIDATION
SET salary = 40000
WHERE emp_id = 1;
```

SELECT * FROM EMPLOYEE_VALIDATION;

Output:

```
+--------+----------+----------+
| emp_id | emp_name | salary   |
+--------+----------+----------+
|      1 | Alice    | 40000.00 |
+--------+----------+----------+
```

Program 4: AFTER UPDATE Trigger to Record Employee Updates

Query/Input:

```
CREATE TABLE EMPLOYEE_UPDATE_AUDIT (
  audit_id INT AUTO_INCREMENT PRIMARY KEY,
  emp_id INT,
  old_salary DECIMAL(10,2),
  new_salary DECIMAL(10,2),
  action_time TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

DELIMITER //

CREATE TRIGGER AfterEmployeeUpdate
AFTER UPDATE ON EMPLOYEE_VALIDATION
FOR EACH ROW
BEGIN
  INSERT INTO EMPLOYEE_UPDATE_AUDIT
    (emp_id, old_salary, new_salary)
  VALUES
    (OLD.emp_id, OLD.salary, NEW.salary);
END //
```

```
DELIMITER ;

UPDATE EMPLOYEE_VALIDATION
SET salary = 45000
WHERE emp_id = 1;

SELECT * FROM EMPLOYEE_UPDATE_AUDIT;
```

Output:

audit_id	emp_id	old_salary	new_salary	action_time
1	1	30000.00	45000.00	2026-10-09 07:36:42

Program 5: BEFORE DELETE Trigger

Query/Input:

```
CREATE TABLE EMPLOYEE_VALIDATION (
   emp_id INT PRIMARY KEY,
   emp_name VARCHAR(50),
   salary DECIMAL(10,2)
);

INSERT INTO EMPLOYEE_VALIDATION VALUES
(1, 'Alice', 30000),
(2, 'Bob', 50000),
(3, 'Charlie', 35000);

DELIMITER //
```

```
CREATE TRIGGER BeforeEmployeeDelete
BEFORE DELETE ON EMPLOYEE_VALIDATION
FOR EACH ROW
BEGIN
  IF OLD.salary > 40000 THEN
    SIGNAL SQLSTATE '45000'
    SET MESSAGE_TEXT = 'Cannot delete employees with salary above 40000';
  END IF;
END //

DELIMITER ;

DELETE FROM EMPLOYEE_VALIDATION
WHERE emp_id = 1;

SELECT * FROM EMPLOYEE_VALIDATION;
```

Output:

```
+--------+----------+----------+
| emp_id | emp_name | salary   |
+--------+----------+----------+
|      2 | Bob      | 50000.00 |
|      3 | Charlie  | 35000.00 |
+--------+----------+----------+
```

Program 6: AFTER DELETE Trigger

Query/Input:

```
CREATE TABLE EMPLOYEE_VALIDATION (
  emp_id INT PRIMARY KEY,
  emp_name VARCHAR(50),
```

```
  salary DECIMAL(10,2)
);

INSERT INTO EMPLOYEE_VALIDATION VALUES
(1, 'Alice', 30000),
(2, 'Bob', 50000),
(3, 'Charlie', 35000);

CREATE TABLE EMPLOYEE_DELETE_AUDIT (
  audit_id INT AUTO_INCREMENT PRIMARY KEY,
  emp_id INT,
  emp_name VARCHAR(50),
  deleted_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

DELIMITER //

CREATE TRIGGER AfterEmployeeDelete
AFTER DELETE ON EMPLOYEE_VALIDATION
FOR EACH ROW
BEGIN
  INSERT INTO EMPLOYEE_DELETE_AUDIT (emp_id, emp_name)
  VALUES (OLD.emp_id, OLD.emp_name);
END //

DELIMITER ;

DELETE FROM EMPLOYEE_VALIDATION
```

WHERE emp_id = 1;

SELECT * FROM EMPLOYEE_DELETE_AUDIT;

Output:

```
+----------+--------+----------+---------------------+
| audit_id | emp_id | emp_name | deleted_at          |
+----------+--------+----------+---------------------+
|        1 |      1 | Alice    | 2026-10-09 07:38:43 |
+----------+--------+----------+---------------------+
```